Prediction of Heart Disease from Clinical Features
Project Report

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Repository: https://github.com/qLessqndr/Prediction-of-Heart-Disease-from-Clinical-Features

1 Introduction

Our goal is binary classification: we predict heart disease with `target_binary = (num > 0)`. We focus on the presence or not of the disease. Part A uses Cleveland alone. Part B pools all sites so we can see how well a model travels when prevalence, missing values, and severity mix change between hospitals.

2 Cleveland baseline

For preprocessing we used median imputation and scaling on `age`, `trestbps`, `chol`, `thalach`, `oldpeak`. For the categorical fields `sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `ca`, `thal` we used most-frequent imputation and one-hot encoding. The train/test split is 80/20, stratified. We compared Naive Bayes, logistic regression, a decision tree (max depth 5), random forest (500 trees), gradient boosting, RBF SVM (C=2), an MLP $(64, 32)$, and a stacking model (LR + RF + SVM with a logistic meta-model and inner five-fold CV).

Table 1: Cleveland holdout results

Model	Test Accuracy	Test F1	Test AUC
Logistic Regression	0.885	0.881	0.966
SVM RBF	0.902	0.897	0.964
Stacking Ensemble	0.869	0.867	0.963
Decision Tree	0.738	0.724	0.745

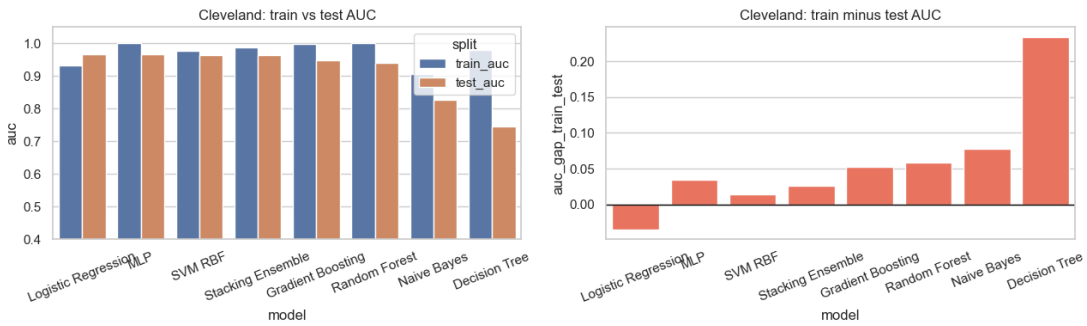

Figure 1: Cleveland benchmark: train vs test AUC (left) and the gap between train and test AUC (right).

Logistic regression gets the best **test AUC (0.966)**, so we treat it as our main ranking score on Cleveland. At the same time, **SVM RBF** has the best **test accuracy (0.902)** and **test F1 (0.897)** on the same split. For logistic regression the classification report gives overall accuracy 0.885 and F1 about 0.889 for class 0 and 0.881 for class 1. The right plot shows a large train–test gap for the decision tree (about 0.234), which looks like overfitting.

Contribution: Alessandro Carnio implemented Cleveland preprocessing and benchmarks; Federico Panico verified splits and overfitting diagnostics.

3 Multicentre data

After merging we have 920 patients. Missing values are not the same everywhere: Cleveland has only 6 missing cells over the 14 inputs; Hungarian has 782; Switzerland 273; VA 698. Positive rates for our binary label also differ a lot: Cleveland 0.459, Hungarian 0.361, Switzerland 0.935, VA 0.745.

Figure 2: Sample size, total missing values, and binary positive rate for each hospital.

Contribution: Alessandro Carnio merged datasets and pipelines; Federico Panico summarised heterogeneity.

4 Models on all sites and validation

We used the same preprocessing as on Cleveland, plus one-hot encoding for `site`. The models are logistic regression, random forest (500 trees), gradient boosting, RBF SVM (C=2), and MLP (64, 32). We evaluated them in three ways: (i) a stratified 80/20 holdout on the pooled table; (ii) GroupKFold with `groups = site` and K=4; (iii) leave-one-site-out with mean, standard deviation, and minimum for AUC and F1.

Table 2: Pooled holdout on merged data

Model	Accuracy	F1	AUC	Brier
Random Forest	0.842	0.860	0.928	0.107
Logistic Regression	0.842	0.863	0.926	0.107
Gradient Boosting	0.859	0.877	0.921	0.106
SVM RBF	0.853	0.874	0.914	0.108
MLP	0.832	0.852	0.889	0.157

Table 3: GroupKFold by site (K=4)

Model	Mean AUC	Std AUC	Mean F1	Mean Acc
Logistic Regression	0.801	0.077	0.764	0.716
SVM RBF	0.793	0.085	0.810	0.779
Random Forest	0.791	0.078	0.770	0.723
Gradient Boosting	0.781	0.067	0.736	0.697
MLP	0.745	0.086	0.749	0.707

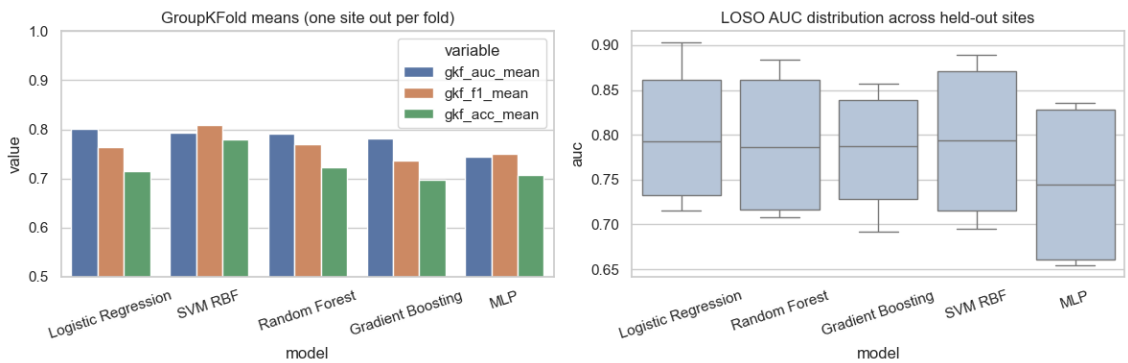

Figure 3: Mean metrics under GroupKFold and LOSO AUC by held-out site.

On the pooled holdout, the best models reach test AUC around 0.89–0.93. GroupKFold is harder: mean AUC goes from **0.745** for MLP up to **0.801** for logistic regression. If we look at mean F1 and mean accuracy across folds, **SVM** is best (**0.810** and **0.779**); logistic regression still wins on mean AUC.

5 LOSO summary, calibration, subgroup errors

Table 4: Leave-one-site-out aggregate

Model	AUC mean	AUC std	AUC min	F1 mean	F1 min
Logistic Regression	0.801	0.089	0.716	0.764	0.728
SVM RBF	0.793	0.099	0.695	0.810	0.768
Random Forest	0.791	0.090	0.708	0.770	0.739
Gradient Boosting	0.781	0.077	0.692	0.736	0.721
MLP	0.745	0.099	0.654	0.749	0.700

For calibration we took Logistic Regression (best mean AUC in the LOSO table) and plotted out-of-fold predicted probabilities from GroupKFold. The **Brier score** is **0.1773**; this metric measures how close predicted probabilities are to the true outcomes, where lower values indicate better calibrated predictions. If the curve follows the diagonal, predicted probabilities are roughly honest; large gaps would mean we should not trust raw scores for risk.

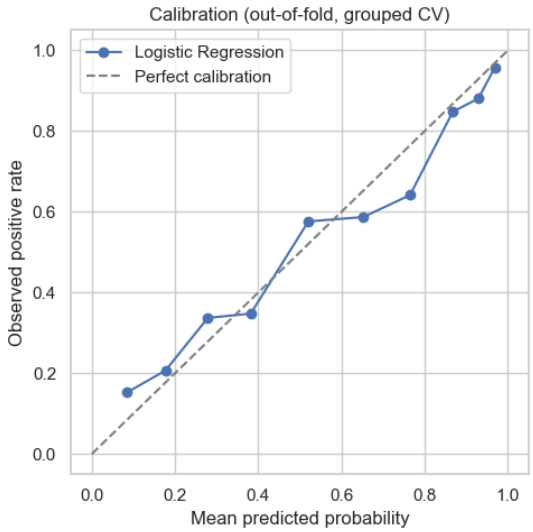

Figure 4: Calibration curve for logistic regression (out-of-fold, grouped CV).

We also grouped errors by `age_group` and `sex`. Misclassification is not uniform: for example females in 56–65 (`sex`=0) reach about 0.21 error rate, and males in 46–55 (`sex`=1) about 0.18. Young females (≤ 45) have much lower error in our table.

Contribution: Alessandro Carnio LOSO and calibration; Federico Panico subgroup error analysis.

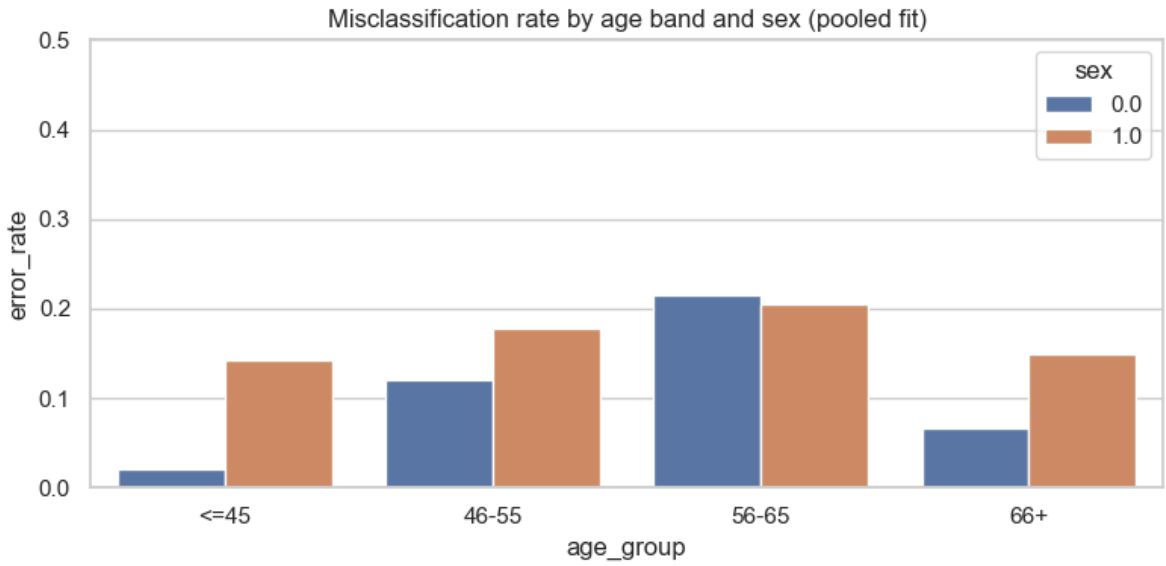

Figure 5: Misclassification rate by age band and sex (pooled fit on all sites; `sex` 0 = female, 1 = male).

6 Conclusion

Table 5: Summary of main results

Setting	Best model (criterion)	Key number	Short comment
Cleveland hold-out	Logistic regression (test AUC)	AUC 0.966	Best ROC ranking on the Cleveland split we used.
Cleveland hold-out	SVM RBF (Acc / F1)	Acc 0.902, F1 0.897	Same split; better accuracy and F1 than logistic regression here.
Merged GroupK-Fold	Logistic regression (mean AUC)	0.801	Best when we validate by leaving out whole sites.
Merged GroupK-Fold	SVM RBF (mean F1, mean Acc)	F1 0.810, Acc 0.779	Higher mean F1 than LR (0.764) across folds.
Merged LOSO	SVM RBF (mean F1)	F1 mean 0.810	Same pattern as in GroupKFold for F1.

Pooled random splits look optimistic compared with GroupKFold and LOSO, which is what we expected when hospitals differ. Logistic regression stays strong on mean AUC when sites move; SVM often wins mean F1 and mean accuracy in grouped validation. Calibration for logistic regression is moderate (Brier 0.1773). In real use we would care about fairness across age and sex, not only average metrics.